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Universidade do Minho
Escola de Engenharia
Departamento de Informática

Diffusion & Multimodality

MESTRADO EM INTELIGÊNCIA ARTIFICIAL

Inteligência Artificial Generativa

2026/2027

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Warm-up: real or generated?

Think of the last AI image, voice or video you saw. Two quick questions for your neighbour:

What gave it away or didn't?
Do you think the model built it left-to-right, like text?



Images

Hands, text, reflections, symmetry



Voice

Prosody, breaths, background



Video

Temporal glitches, physics

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Part 1 · Diffusion

Generate by turning noise into signal, step by step

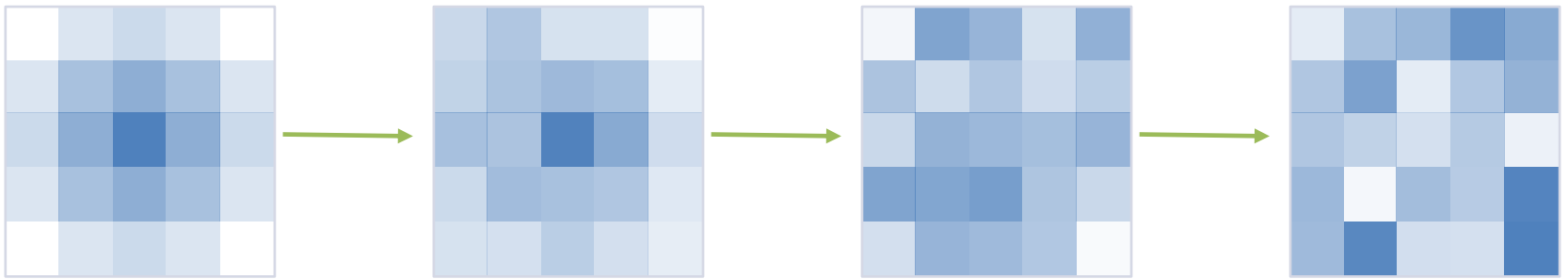
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Four ways to model a distribution

GANs	VAEs	Autoregressive	Diffusion
A generator fools a discriminator	Encode to a latent, decode back	Predict the next token/pixel	Denoise noise into data
Sharp, fast; unstable, mode collapse	*Stable latent; often blurry*	*Great for text; slow for images*	*High quality, controllable; many steps*

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Forward: add noise.
Reverse: learn to remove it



forward $q(x_t \mid x_{t-1})$: add a little Gaussian noise →

← **reverse $p\theta(x_{t-1} \mid x_t)$: a network predicts and removes the noise**

The forward process is fixed and needs no learning. All the learning is in the reverse process.

Sohl-Dickstein 2015; Ho et al. 2020, "Denoising Diffusion Probabilistic Models"

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Training: just predict the noise

1. Take a real image x_0 and a random step t
2. Add t steps of noise to get a noisy x_t (one-shot formula)
3. Ask the network $\varepsilon\theta(x_t, t)$ to predict the noise that was added
4. Loss = $\|\varepsilon - \varepsilon\theta(x_t, t)\|^2$ · backprop · repeat

Why it works

- A simple regression target, not an adversarial game — stable to train.
- Predicting noise ≈ pointing toward the data (the score).
- One network handles every noise level, shared across t.

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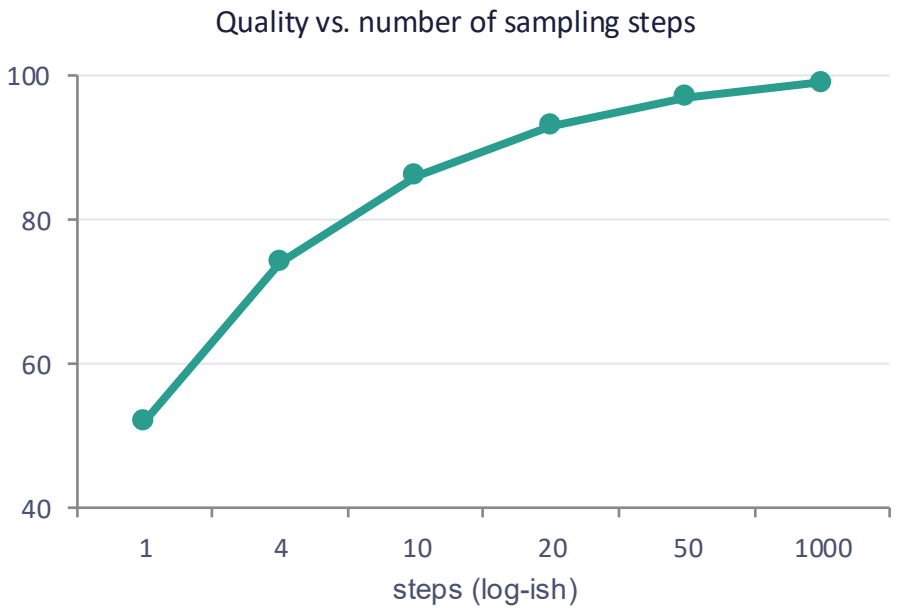
Sampling: the quality–speed dial

DDPM

- Stochastic; ~1000 tiny steps
- High quality, slow

DDIM & fast solvers

Deterministic; 20–50 steps for similar quality.
Distillation pushes it to 1–4 steps.



Illustrative; diminishing returns after a few dozen steps

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Concept check: why does DDIM work?

VOTE

You trained the model with **200** noising steps. At generation, DDIM makes clean moons in only **20** steps. Why?

A The model memorised 20 specific images during training.

B DDIM takes larger, near-deterministic steps along a smoother path.

C 20 steps add more noise than 200, which sharpens the result.

D It only works because the data is 2-D.

Vote → convince your neighbour (90s) → **vote again**

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Steering generation: text and guidance

Conditioning

- Feed a text embedding (e.g. from CLIP/T5) into the denoiser via cross-attention.
- Also condition on images, poses, depth, masks (ControlNet, inpainting).

Classifier-free guidance

$$\varepsilon_x = \varepsilon(\emptyset) + w \cdot (\varepsilon(\text{prompt}) - \varepsilon(\emptyset))$$

- Run the denoiser with and without the prompt; push away from the unconditioned direction.
- Guidance scale w: higher = more prompt-faithful, less diverse, can look fried.

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Live demo: turn the guidan

LIVE DEMO

Predict first: at guidance w = 15, do the samples spread across the whole moon, or collapse to a dot?

```
# in the lab notebook, Part 4
for w in [0, 1, 3, 8, 15]:
    s = guided_sample(cmodel,
        label=0, w=w)
    plot(s)
```

Watch for

w = 0	samples fill BOTH moons — unconditioned
w = 1	the labelled moon only
w = 3	tight, faithful to the class
w = 15	collapsed, over-saturated — the ‘fried’ look

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Latent diffusion: do it in a smaller space



A 512×512×3 image ≈ 64×64×4 latent: ~48× fewer values to denoise.

Why it mattered

Latent diffusion (Stable Diffusion, 2022) made high-res generation runnable on a single consumer GPU.

U-Net → DiT

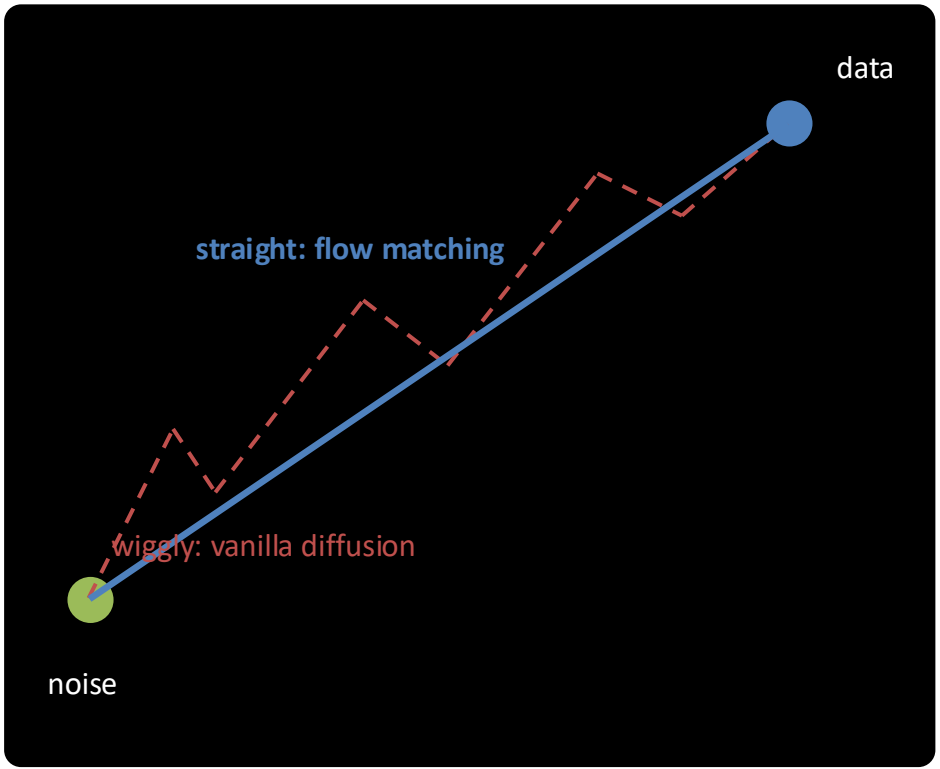
The denoiser was a U-Net; newer models use a Diffusion Transformer, which scales better (SD3, Flux, Sora).

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The modern recipe: flow matching

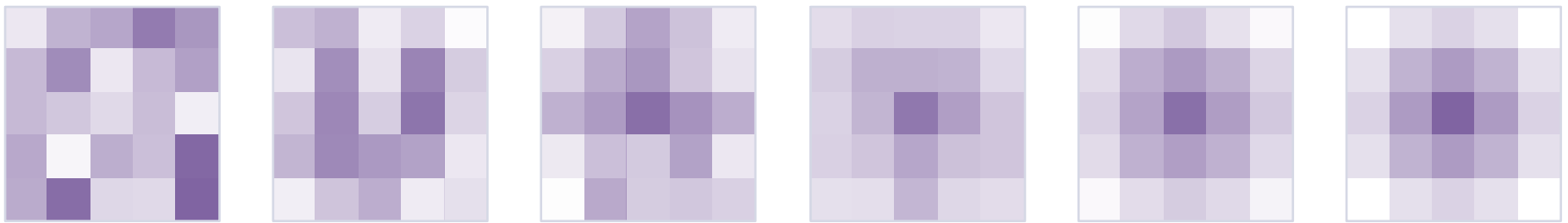
The idea

- Learn a velocity field that moves points from noise to data along (nearly) straight paths.
- Train by regressing that velocity — same spirit as denoising, cleaner maths.
- Straighter paths ⇒ fewer steps to sample.



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Example: generating one image



prompt: "a red bicycle leaning on a white wall", guidance scale 7

Start

A latent of pure Gaussian noise, plus the text embedding.

Each step

The denoiser predicts the noise; a slice is removed; the prompt steers via guidance.

End

The clean latent is decoded by the VAE into pixels.

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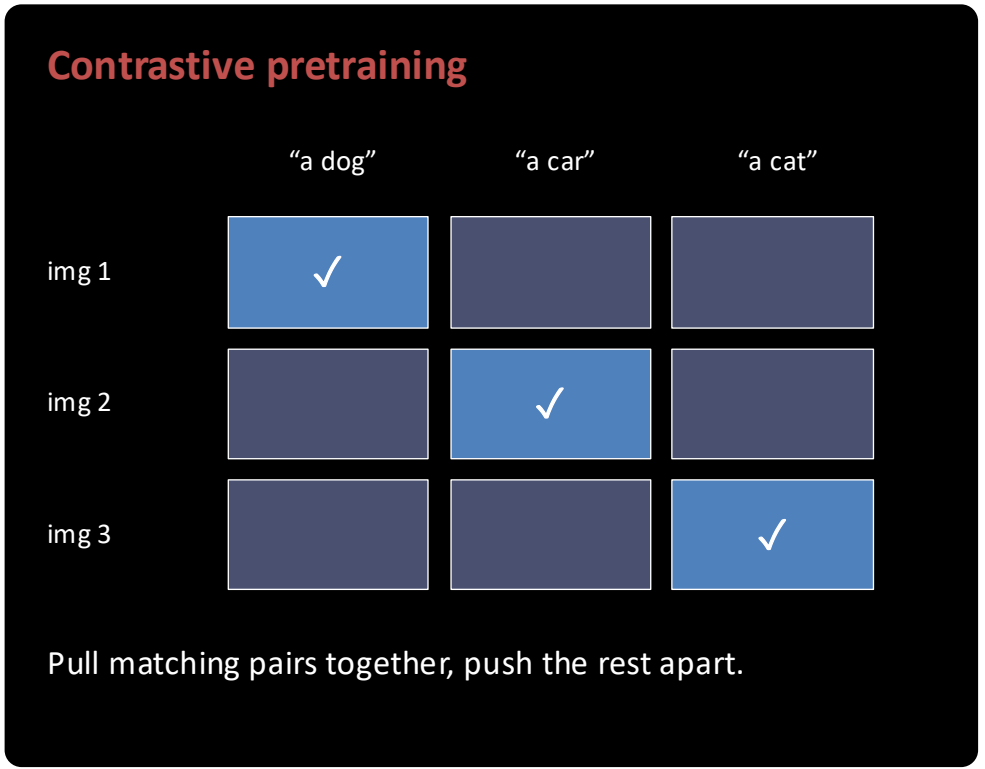


Part 2 · Multimodality

One model that can see, hear and speak

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CLIP: aligning images and text



Two encoders

An image encoder and a text encoder map to one shared space.

400M+ pairs

Trained on huge image–caption data scraped from the web.

Zero-shot

Classify by comparing an image to label texts — no fine-tuning.

The bridge

SigLIP and friends are the ‘eyes’ bolted onto today’s VLMs.

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How to give an LLM eyes (a VLM)

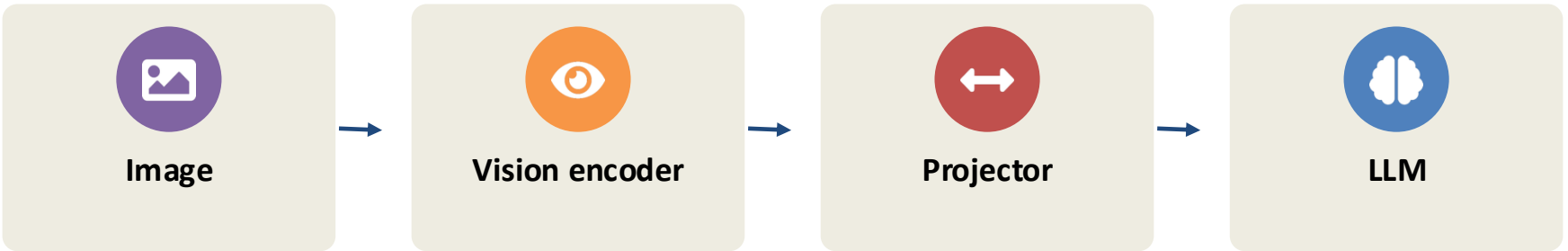


Image → patch embeddings → projected into the LLM's token space → read alongside text tokens

Vision encoder

A frozen CLIP/SigLIP turns the image into patch embeddings.

Projector

A small MLP maps those into the LLM's embedding space, as 'image tokens'.

LLM

Attends to image and text tokens together to answer, caption, or reason.

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Bolted-on vs. native multimodality

Late fusion (adapter)

- Pretrained vision encoder + pretrained LLM, joined by a projector.
- Cheap, modular, fast to build.
- Vision is an add-on; limited for dense visual reasoning.

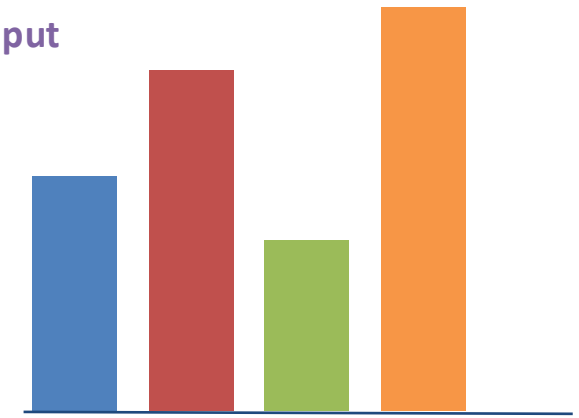
Early / native fusion

- Trained on mixed modalities from the start; one token stream for all.
- Stronger cross-modal reasoning.
- Far more expensive to train; data mixing is hard.

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Example: a VLM reads a chart

Input



“Which quarter was highest, and by how much over Q3?”

1. **Encode**
   The chart becomes ~256 patch embeddings → image tokens.
2. **Attend**
   The LLM aligns ‘Q4’ token with the tallest bar’s patches.
3. **Reason**
   Reads bar heights, subtracts, and composes an answer.
4. **Answer**
   “Q4 is highest, about 2.4× the height of Q3.”

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Concept check: can CLIP count?

For a photo of **two dogs**, CLIP scores “a photo of a dog” and “a photo of two dogs” almost equally. The best explanation is…

A. CLIP cannot actually see the image.

B. Contrastive training optimises global image–text match, not counting.

C. The temperature is set too low.

D. The image encoder is broken and needs fine-tuning.

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Video: add the time axis

Time is expensive

A clip is many frames; naive attention is quadratic in space × time.

Space-time patches

Latent video models tokenise 3D ‘tubelets’ and run a DiT over them.

Consistency is hard

Objects must persist; physics must hold across frames.

Where it is

Short, striking clips; length, control and coherence still improving.

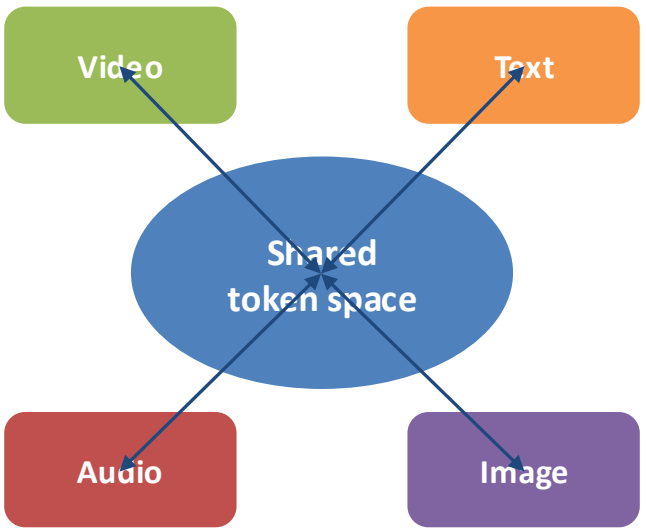
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Audio: understanding and generating sound

Speech-to-text	Whisper-style encoder-decoders transcribe and translate robustly.
Text-to-audio	Diffusion or token models generate speech, music and effects.
Representations	Mel-spectrograms, or discrete audio tokens from a neural codec.
Voice agents	Streaming speech-to-speech enables low-latency spoken assistants.

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Any-to-any: one model, many modalities



Tokenise everything

Each modality becomes discrete tokens or latents in a common space.

Mix in and out

Input any combination; the model can emit any combination.

Emerging

Unified understanding + generation is the current frontier (e.g. GPT-4o-style, Gemini).

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Generative media: risks and provenance



Deepfakes & fraud

Voice clones, fake video, non-consensual imagery, scams at scale.



Hard to evaluate

FID/CLIPScore are weak; quality is subjective and human eval is costly.



Provenance

C2PA / Content Credentials sign origin; invisible watermarks (SynthID).



Rules

EU AI Act: label AI-generated and deepfake content (transparency duties).

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Exit ticket

1. In diffusion training, what does the network actually learn to predict?
2. Why run diffusion in a VAE's latent space instead of on pixels?
3. What does classifier-free guidance control?
4. In a typical VLM, what turns an image into something the LLM can read?
5. Name one way to mark that media was AI-generated.

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Five things to remember

Diffusion = denoise noise	Learn to remove noise step by step; predict the noise.
Work in latent space	VAE + denoiser (U-Net → DiT); flow matching straightens the path.
Guidance steers it	Text conditioning + classifier-free guidance trade fidelity for diversity.
VLMs = encoder + LLM	CLIP-style eyes projected into an LLM's token space.
Provenance over detection	Sign origin (C2PA, watermarks); labelling is now a legal duty.

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References & further reading

Diffusion

Ho et al. (2020)
Denoising Diffusion Probabilistic Models (DDPM)

Song et al. (2021)
DDIM: faster deterministic sampling

Ho & Salimans (2022)
Classifier-Free Guidance

Rombach et al. (2022)
Latent Diffusion / Stable Diffusion

Peebles & Xie (2023)
Scalable Diffusion Models with Transformers (DiT)

Lipman et al. (2022)
Flow Matching for Generative Modeling

Multimodality

Radford et al. (2021)
CLIP: learning transferable visual models

Zhai et al. (2023)
SigLIP: sigmoid loss for image–text

Liu et al. (2023)
LLaVA: visual instruction tuning

Radford et al. (2022)
Whisper: robust speech recognition

OpenAI (2024)
Video generation as world simulators (Sora)

EU (2024)
AI Act: transparency for generated content

This field moves monthly — check each project's current page before citing figures.

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